

Theme – Agriculture, FoodTech & Rural Innovation

Title – Direct Farmer-to-Consumer Digital Marketplace Platform

Category- Software

Team Name – Kishan Crew

Khet se Kitchen

Idea / Solution

Khetse is a **user-friendly, AI-enabled digital marketplace** that connects farmer directly with consumer, ensuring transparent pricing, improved income, and efficient farm-to-home delivery

- Platform connecting farmer with consumer (with no intermediate)
- Fair price comparison which ensures transparency & costumer trust
- Multi-role integrated platform :-
  - Farmer – list product, manage inventory
  - Consumer – browse, search & order
  - Admin – dashboard & analytics
  - Delivery - order tracking system
- Helps farmers to earn more and support them with organic farming
- Focus on fresh & local product

Problem Resolution

- While several platform address isolated aspects of the problem. Our solution integrates marketplace, transparency, AI guidance, and user-friendly UI into a single unified system.
- Eliminates middlemen that directly beneficial for both farmers and consumer

Unique Value Proposition (UVP)

- Authentication for every user (farmer & consumer)
- Authentication done in less than a minute
- Platform provides multilingual supports for user
- Provides AI based search box and chatbot
- Platform support voice command

SOLUTION APPROACH

Algorithm Development:
Node.js & Express – Core technologies used for developing the platform.

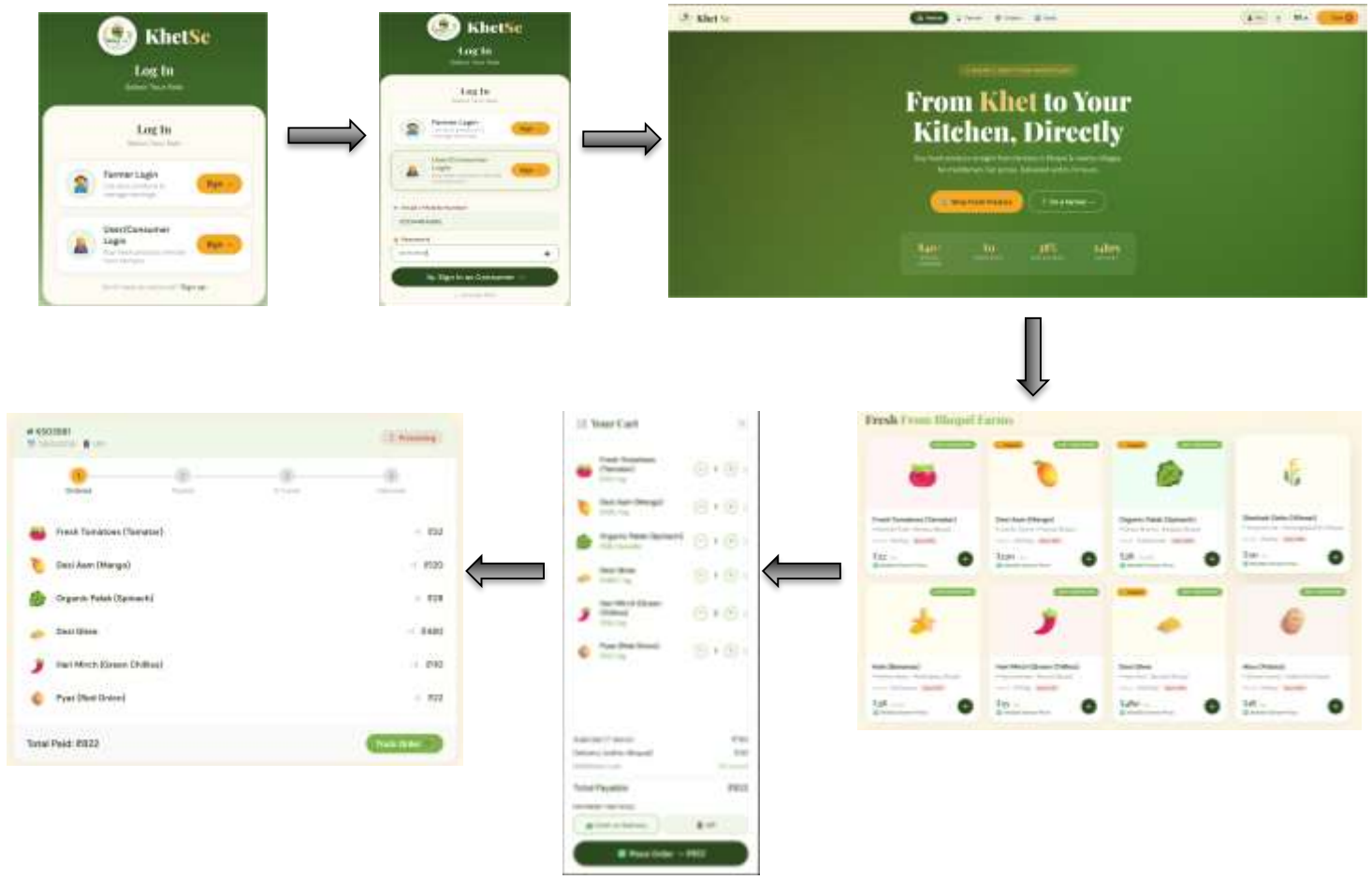
Mobile Application Development:
Flutter - Framework used for building the mobile application, ensuring cross-platform compatibility.

Encryption and Security:
Custom encryption algorithm powered by 8 layer octa 512 bit encryption for secure data transmission and authentication

Database Management Services:
Mongodb – Relational database management

UI/UX:
HTML, CSS, Javascript

Process flow architecture:



Product Status: 70% product built completed and further build is on process. Testing and validation process are next to be undergone.

FEASIBILITY AND VIABILITY

Technical Feasibility – 85%
- ✓ Frontend-Flutter(Mobile App), HTML/CSS/JS
- ✓ Backend-Node.js, Express.js
- ✓ Database-Mongodb
- ✓ APIs-REST APIs
- ✓ Authentication-Firebase Auth / JWT
- ✓ Cloud-Firebase / AWS
- ✓ Payment-RazorPay
- ✓ Notifications-Firebase Cloud Messaging
- ✓ Maps-Google Maps APIs

Operational Feasibility –
- ✓ Simple UI (icons, Hindi toggle, voice input)
- ✓ Farmer- friendly form (easy listing)
- ✓ Local delivery model

Revenue models:
- ✓ Commission (2-5% per order)
- ✓ Subscription for farmers (premium listing)
- ✓ Delivery charges
- ✓ Ads (agri products, fertilizers)

S. No.	Challenge	Description	Proposed Solution	Strategy Insight
1	Farmer Adoption	Farmers may lack trust in digital platforms and have low technical skills.	Awareness campaigns, FPO partnerships, voice & Hindi support, field agents	Offline trust → Online adoption
2	Logistics & Delivery	Managing last-mile delivery, perishables & timely delivery.	Hyperlocal model, delivery partners, route optimization, cluster deliveries	Control geography first
3	Quality Assurance & Trust	Consumers unsure about freshness, quality & authenticity.	Farmer verification, ratings & reviews, product details, quality checks	Build digital trust signals
4	Price Fluctuation & Market Competition	Prices change daily; competition from mandis & other platforms.	Dynamic pricing, mandi vs farmer price comparison, AI price prediction	Transparency = Competitive advantage
5	Payment & Financial Challenges	Farmers prefer cash; risk of failed transactions; delayed settlements.	COD + UPI options, secure payments, fast & assured settlements	Flexibility builds trust
6	Technology Barriers	Low digital literacy, poor internet connectivity in rural areas.	Simple UI, voice input, Hindi support, offline mode & low data usage	Design for rural users
7	Supply-Demand Imbalance	Overproduction leads to wastage; shortages lead to unsatisfied buyers.	AI demand prediction, pre-orders, real-time alerts to farmers	Data-driven decisions
8	Competition from Existing Platforms	Strong competition from BigBasket, DeHaat, local vendors, etc.	Focus on differentiation (AI assistant, transparency, local model, fair pricing)	Win locally before scaling
9	Legal & Regulatory Compliance	Need to comply with GST, FSSAI, e-commerce & data regulations.	Proper registration, follow guidelines, automated invoicing & compliance	Compliance early = No future risk
10	Farmer Retention	Farmers may leave due to low sales, delayed payments or less support.	Analytics dashboard, incentives, training, fast payouts & support	Make farmers feel valued & supported
11	Customer Retention	Users may switch to local vendors or cheaper platforms.	Loyalty programs, subscriptions, personalized offers, excellent UX	Convenience + trust = retention
12	Scaling Challenges	Expanding to new cities brings logistics & operational complexity.	City-wise expansion, local partnerships, strong backend & modular system	Scale in modules, not all at once

OUR APPROACH OUR STRENGTH | 12+ Key Challenges Identified | 12+ Practical & Scalable Solutions | 100% Focus on Farmer & Customer Value | Long-Term Sustainable Growth | Strong Local Start, Scalable National Impact

Feature	Existing Apps	Our Prototype	Advantage Score
Direct farmer → consumer	Partial	Full	95%
Price transparency (mandi vs farmer)	Rare	Strong feature	90%
AI farmer assistant	Very limited	Included	92%
Voice + Hindi support	Partial	Strong	88%
Full UI flow (cart, orders, tracking)	Partial	Complete	96%
Local marketplace model	Some	Clearly implemented	90%

Overall Advantage of Our Prototype | 6/6 Features Superior | 100% Feature Coverage | ~92% Overall Advantage Score

IMPACT AND BENEFITS

Farmers (Primary Target Audience)

Economic Impact:

- Higher profit margins (no middlemen cuts)
- Direct control over pricing
- Faster and more reliable payments

Growth Impact:

- Access to a larger digital market
- Ability to sell beyond local mandis
- Better demand visibility

Behavioral Impact:

- Shift from traditional selling > digital selling
- Increased confidence & independence

Consumers (Urban & Semi-Urban)

Financial Impact:

- Lower prices compared to retail/mandi
- Better value for money

Quality Impact:

- Fresher & farm-direct produce
- Transparency of source

Convenience Impact:

- Easy ordering, tracking, delivery
- Digital payment + COD

Trust Impact:

- See farmer details & product origin
- Transparent pricing comparison

Local Vendors & Retailers

Business Impact:

- Direct sourcing from farmer
- Reduce procurement cost

Operational Impact:

- Stable supply chain
- Better price negotiation

Delivery Partners / Local Workforce

Employment Impact:

Job creation:

- Delivery agent
- Logistic coordinators

Local opportunities:

- Work within local region

Government & Policy Makers

Economic Impact:

- Strengthens agriculture economy
- Improves farmer income level

Development Impact:

- Supports digital India & AgriTech Initiatives
- Encourages rural digitization

System Impact:

- Reduce dependency on tradition mandi system

RESEARCH AND REFERENCES

- krishify.com/farmer-app/
https://share.google/XFh2M1qjmjogpsBZl

- Kisaan.net | Sugarcane Farmers
https://share.google/CPyiU1cROvAK5XZ1p

- fresh-zone-reveal
https://share.google/FPFcvqmWhXviNOts1